

THE FROBENIUS EIGENVALUES OF ARTIN-SCHREIER-WITT CURVES ARE GAUSS SUMS

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ABSTRACT. We may geometrically diagonalize Frobenius to compute its trace in order to generalize the Davenport-Hasse theorem and show that eigenvalues of Frobenius of Artin-Schrier-Witt covers are also Gauss sums. We propose this as a method of recovering the slopes of their Jacobians.

1. (BRIEF) BACKGROUND

I've felt for a long time that automorphisms of curves should control or at least exert serious force on the slopes on their Jacobians. This playful note is toward exploring this force, and outlining a possible method of exploiting it.

Correspondences on X are endomorphisms of X . Given $Y \subseteq X \times X$, the associated correspondence gives an endomorphism of X via $p_2(p_1^{-1}(X))$, if the divisor Y we pick doesn't have weird zeros (so that it doesn't just erase all of X).

$$\begin{array}{ccc}
 & Y & \\
 & \downarrow & \\
 & X \times X & \\
 \swarrow p_1 & & \searrow p_2 \\
 X & & X
 \end{array}$$

Given a map $X \xrightarrow{\Delta} X \times X$, take the Frobenius map:

$$X \xrightarrow{F} X \times X$$

$$\Delta X \cap FX = \#X(\mathbb{F}_p)$$

If $Y \simeq FX$ is linearly equivalent, then,

$$\#(Y \cap \Delta X) = \#X(\mathbb{F}_q).$$

Inspired by the beautiful proof of the Lefschetz fixed point theorem via correspondences.

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2. REPROOF OF DAVENPORT-HASSE GEOMETRICALLY DIAGONALIZING

Let k be a field of characteristic p , such that $K \supseteq F_q$. We begin with an Artin-Schreier curve A with affine equation

$$y^p - y = x^{p-1}$$

Remark. Coleman considers the more general case of $y^p - y = x^{p^f-1}$, we do not consider p^f here (yet!), as the x terms of the divisor g_P are more complicated for the p^f case and I'm just trying to do a sanity check.

Consider the following homomorphisms:

$$\begin{aligned} \psi : \mathbb{F}_q &\rightarrow \text{Aut}(A) \\ a &\mapsto \psi(a)(x, y, z) = (x, y + a) \\ \chi : \mathbb{F}_q^\times &\rightarrow \text{Aut}(A) \\ b &\mapsto \chi(b)(x, y, z) = (b^{(q-1)/m}x, y) \end{aligned}$$

Definition 2.1. *We consider the multiplication to be within the group structure of $\text{Aut}(A)$, and the addition to be the addition of cycles.*

We then define the following automorphism,

$$G := - \sum_{b \in \mathbb{F}_q} \chi(b) \psi(b),$$

and consider its associated correspondence G .

We also conflate the notation to also call the divisors associated to these correspondences by the same name.

Here is an excellent source on additive and multiplicative characters to get a feel for them if they are new friends for you. <https://kconrad.math.uconn.edu/blurbs/gradnumthy/Gauss-Jacobi-sums.pdf>

Theorem 2.2. *(1) Under equivalence of correspondences, $\phi = G$.*

Proof. We will construct a principal divisor $(g_P) \in \text{Div}(A)$ and show it agrees with the divisor $(\phi - G)(P - \infty) := \phi(P) - G(P) - \phi(\infty) + G(\infty)$ on all but finitely many P . Given a point P , a finite point on A over k such that $x(P) \neq 0$, let

$$g_P(Q) := y(Q) - y(P) - \frac{x(Q)}{x(P)}.$$

varying Q ,

We have reduced the claim to showing

$$(g_P) = (\phi - G)(P - \infty).$$

We will now calculate zeros and poles of g_P and $(\phi - G)(P - \infty)$ to show that they coincide, thus establishing this equivalence.

We mention this for our pole computations. The point at infinity of A is $\infty := [1 : 0 : 0]$, and our uniformizers are as follows.

$$u_P = \begin{cases} y - y(P) & \text{if } y(P) \neq 0 \text{ and } P \neq [1 : 0 : 0] \\ x & \text{if } x(P) = 0 \\ \frac{y}{x} & \text{if } P = [1 : 0 : 0] \end{cases}$$

This divisor (g_P) has p zeros, it vanishes at

- (A) the $p - 1$ points $(\chi(b)\psi(b))(x(P), y(P))$, $b \in \mathbb{F}_q^*$

$$\begin{aligned} g_P(\chi(b)\psi(b)(x(P), y(P))) &= g_P(bx(P), y(P) + b) \\ &= (y(P) + b) - y(P) - \frac{bx(P)}{x(P)} = 0 \end{aligned}$$

- (B) at the point $(x(P)^p, y(P)^p)$ which is distinct from the (A) points if $x(P)^{p-1} \notin \mathbb{F}_p^\times$.

$$\begin{aligned} g_P(\phi(P)) &= g_P(x(P)^p, y(P)^p) \\ &= y(P)^p - y(P) - \frac{x(P)^p}{x(P)} = 0 \end{aligned}$$

By Lemma 2.3, we have only one pole at ∞ of order p . Putting it all together, we get:

$$\begin{aligned} \text{div}(g_P) &= \text{div}_0 g_P - \text{div}_\infty g_P \\ &= \left(\sum_{b=1}^{p-1} [\chi(b)\psi(b)(x(P), y(P))] + [(x(P)^p, y(P)^p)] \right) - p[\infty] \end{aligned}$$

By Lemma 2.4, the divisor $(\phi - G)(P - \infty)$ has the same form.

Thus, we have shown that the divisors (g_P) and $(\phi - G)(P - \infty)$ have the same zeros and poles. The divisors being linearly equivalent gives us an equivalence of their associated correspondences.

□

Lemma 2.3. *(g_P) has a pole at ∞ of order p and no other poles.*

Proof. We mention this for our pole computations: The point at infinity is $[1 : 0 : 0]$. The uniformizer of the curve at infinity is $T := \frac{y}{x}$ and elsewhere is y .

At ∞ , the curve is $Y^p - YZ = Z$ where $[Y : Z] := [\frac{y}{x} : \frac{z}{x}]$, we can nest this to find the order of the uniformizer.

$$\begin{aligned} Z &= Y^p - Y(Y^p - YZ^{p-1})^{p-1} \\ &= f(Y) = \mathcal{O}(Y^p). \end{aligned}$$

Thus, there is a pole at infinity with order p . There are no other poles because the curve is totally wildly ramified at infinity (thus there is only one point above the pole infinity in $\mathbb{P}^1$). $\square$

Lemma 2.4. *The divisor*

$$(\phi - G)(P - \infty) = \left(\sum_{b=1}^{p-1} [\chi(b)\psi(b)(x(P), y(P))] + [(x(P)^p, y(P)^p)] \right) - p[\infty].$$

Proof. • $G(\infty) = (p-1)[\infty]$. In other words, $G(\infty)$ takes ∞ and sends it to $p-1$ translates of itself. Let's see why, for a fixed b $\chi(b)\psi(b)([1 : 0 : 0]) = \chi(b)([1 : 0 : 0])\phi(b)([1 : 0 : 0]) = [b : 0 : 0][1 : b : 0] = [b : 0 : 0] = [1 : 0 : 0]$.

- $G(P) = \sum_{b \in \mathbb{F}_p^\times} (bx(P), y(P) + b)$
- $\phi(P) = (x(P)^p, y(P)^p)$
- $\phi(\infty) = -\infty$

In other words, $G(\infty) - \phi(\infty) = p[\infty]$, and $\phi(P) - G(P) = [(x(P)^p, y(P)^p)] - \sum_{b \in \mathbb{F}_p^\times} [\chi(b)\phi(b)(x(P), y(P))]$.

Thus,

$$\begin{aligned} \phi(P) - \phi(\infty) + G(\infty) - G(P) &= [(x(P)^p, y(P)^p)] + [\infty] \\ &\quad + (p-1)[\infty] - \sum_{b \in \mathbb{F}_p^\times} [\chi(b)\psi(b)(x(P), y(P))] \\ &= \left(\sum_{b=1}^{p-1} [\chi(b)\psi(b)(x(P), y(P))] + [(x(P)^p, y(P)^p)] \right) - p[\infty] \end{aligned}$$

$\square$

Corollary 2.5. *The eigenvalues of Frobenius for this Artin-Schreier-Witt curve A are G_q .*

Proof. Recall that $H_{\text{ét}}^1(A)$ is the Divisor class group. Given that the Frobenius map and G_q are equivalences of correspondences, they are also equivalences of operators acting on $H^1(A; \mathbb{Q}_\ell)$. The eigenvalues of an operator on a vector space are eigenvalues on the underlying field. $\square$

Something I have yet to understand is how to explicitly evaluate the eigenvalues of the Gauss sum correspondence on the underlying field explicitly.

3. SLOPE ANALYSIS

Let's review what Manin did for the Artin-Schreier curve to see which explicit eigenvalues we *really* get. We may consider ψ to be a multiplicative character, and χ to be an additive character. Let $t \in \mathbb{F}_p^\times$ and let choose a lift $\tilde{t}$ to $W(k)$.

Let ζ be a p th root of unity. We define $\phi_i(t) = \zeta^{i\tilde{t}}$ where i , and $\chi_j(t) = -\tilde{t}^{-j}$.

Theorem 3.1. *(Thm 4.1 Manin's thesis (2)) The eigenvalues of Frobenius on the curve $X : y^p - y = x^{p-1}$ are sums of the following form.*

$$\tau(\psi_i, \chi_j) = \sum_{t \in \mathbb{F}_p^\times} \phi_i(t) \chi_j(t).$$

where $1 \leq i \leq p-1$ and $1 \leq j \leq p-2$.

Corollary 3.2. *The slopes of the p -divisible group associated to the curve X are $\{1/(p-1), 2/(p-1), \dots, (p-2)/(p-1)\}$.*

Proof. We use the eigenvalues above. The key observation of Stickelberger is that for $\lambda = 1 - \zeta$,

$$\tau(\phi_i, \chi_j) = -j^{-1} \lambda^j \pmod{\lambda^{j+1}}.$$

Since $v_p(\lambda) = 1/(p-1)$, this means that $v_p(\tau(\phi_i, \chi_j)) = \frac{j}{p-1}$. So we get $(p-1)$ copies of each $1 \leq j \leq (p-2)$. Each of these eigenvalues has multiplicity $p-1$. $\square$

Conjecture 3.3. *The slope decomposition of the Dieudonné module associated to the Jacobian of A_k ($\mathbb{Z}/p^k$ totally wildly ramified cover) always has a piece of slope $1/p^{k-1}(p-1)$.*

Some support for the conjecture: I have calculated (thanks to the ATLAS North-western super computer) only pieces of the $p=3$ here $h=p(p-1)=6$ case in support of this, so at the moment the conjecture is quite flimsy.

- $y^3 - y = 1/x^2, w^3 - w = -x^2y^2 - x^4y$ is the minimal genus ASW curve X – genus 16, with slope decomposition $G_{1/6} \times 2G_{1/3} \times 4G_{1/2} \times 2G_{2/3} \times G_{5/6}$
- $y^3 - y = 1/x^2, w^3 - w = -x^2y^2 - x^4y - y^5 - y^7$ is an Artin-Schreier-Witt curve with genus 29, with slope decomposition $6G_0 \times G_{1/6} \times 2G_{1/3} \times 11G_{1/2} \times 2G_{2/3} \times G_{5/6} \times 6G_1$

4. ARTIN-SCHREIER-WITT CASE FOR $\mathbb{Z}/p^2$

We just repeat the above but in greater generality. Slay on his part.

Definition 4.1. *Let*

$$C(z, w) := \frac{z^p + w^p - (z + w)^p}{p},$$

then the equations which define our Artin-Schreier-Witt curve A .

$$\begin{aligned} w^p - w &= C(x^{p-1}, y) \\ y^p - y &= x^{p-1} \end{aligned}$$

This curve is still totally wildly ramified at infinity) and an extension of the curve in the previous section.

Definition 4.2. *Consider the following homomorphisms:*

$$\begin{aligned} \psi : \mathbb{F}_q &\rightarrow \text{Aut}(A) \\ a &\mapsto \psi(a)(x, y, z) = (x, y + a, w - C(a, y)) \\ \chi : \mathbb{F}_q &\rightarrow \text{Aut}(A) \\ b &\mapsto \chi(b)(x, y, z) = (b^{(q-1)/m}x, y, z) \end{aligned}$$

We define addition as

$$\psi : a + b(x, y + (a + b), w - (C(a, y) + C(b, y))).$$

We then define the following automorphism,

$$G := - \sum_{b \in \mathbb{F}_q} \chi(b) \psi(b),$$

where the multiplication is given within the automorphism group, and the addition is via addition of cycles. We may regard G as a correspondence on the Artin-Schreier-Witt curve A .

Remark. Notice that in the above definition, the image of homomorphism ψ is $\mathbb{Z}/p^2$.

Theorem 4.3. *Under equivalence of correspondences, $\phi = G$.*

Proof. We construct a principal divisor g_P such that it agrees with $(\phi - G)(P - \infty)$ on all but finitely many P . Given a point P , a finite point on A over k such that $x(P) \neq 0$, let

$$g_P(Q) := w_Q - w_P - C\left(\frac{x_Q}{x_P}, y_P\right) \in \text{Div}(A).$$

This divisor (g_P) has a pole at ∞ of order p^2 and no other poles, (g_P) also has zeros at the following p^2 points:

- Let us observe first the zeros of $g_P(Q)$ where $Q := \phi(P) := (x_P^p, y_P^p, w_P^p)$, this is then

$$\begin{aligned} g_P(\phi(P)) &= w_P^p - w_P - C\left(\frac{x_P^p}{x_P}, y_P\right) \\ &= w_P^p - w_P - C(x_P^{p-1}, y_P) \\ &= 0 \end{aligned}$$

- We then observe $g_P(\chi(1)\psi(1)) = g_P(x_P, y_P + 1, w_P + C(b, y_P))$,

$$\begin{aligned} g_P(\chi(1)\psi(1)) &= w_P + C(b, y_P) - w_P - C\left(b \frac{x_P}{x_P}, y_P\right) \\ &= C(b, y_P) - C(b, y_P) \\ &= 0. \end{aligned}$$

It is sufficient to then calculate zeros and poles of $(\phi - G)(P - \infty)$, and show it agrees with that of g_P shown above. In other words, Conjecture 4.4.

If we do this, it proves the claim and establishes the theorem since we've shown that $(\phi - G)(P - \infty)$ is principal for all but finitely many P . $\square$

If we could show the only zeros of g_P are the p^2 as above, the only zero of g_P is ∞ and it has order p^2 . We'd be golden. More precisely, it's left to show:

Conjecture 4.4. *(points of desire)*

$$(\phi - G)(P - \infty) = (x(P)^p, y(P)^p) + \sum_{b \in \mathbb{F}_p^\times} \chi(b) \phi(b)(x(P), y(P)) - p^2[\infty]$$

Corollary 4.5. *The eigenvalues of Frobenius for this Artin-Schreier curve A are the eigenvalues of the operator G .*

Proof. Recall that $H_{\text{ét}}^1(A)$ is the Divisor class group. Given that the Frobenius map and G_q are equivalences of correspondences, they are also equivalences of operators acting on $H^1(A; \mathbb{Q}_\ell)$. The eigenvalues of an operator on a vector space are eigenvalues on the underlying field. $\square$

REFERENCES

- [1] Robert Coleman, *On the Frobenius Endomorphisms of Fermat and Artin-Schreier Curves*.
- [2] Yuri Manin, *The Theory of Formal Groups over Fields of Finite Characteristic*.